

Statistical Inference and Multivariate Analysis (MA324)

LECTURE SLIDES
Topic 03
Multiple Linear Regression

Jan-May 2026

Multiple Linear Regression

- In general, the response (y) may be related to p regressors (input variables/predictors).
- The model

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i, i = 1, \dots, n$$

is called **multiple linear regression**. A regression model with p regressors.

- The parameters $\beta_j, j = 0, 1, 2, \dots, p$ are called regression coefficients.
- $\beta_j, j = 0, 1, 2, \dots, p$ represents the change in the average value of the response for a unit change in j^{th} regressor keeping other regressors fixed.

- Data (of sample size n): $(y_i, x_{i1}, x_{i2}, \dots, x_{ip}), i = 1, \dots, n$
- Scalar notation becomes cumbersome – so matrix notation is used
- Let

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, X = \begin{pmatrix} 1 & x_{11} & \dots & x_{1p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{pmatrix}, \beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_p \end{pmatrix}, \epsilon = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}$$

- Then we can write the model in a more compact form:

$$y_{n \times 1} = X_{n \times (p+1)} \beta_{(p+1) \times 1} + \epsilon_{n \times 1}$$

- X is called the ***design matrix***

Error Assumptions

- As before, ϵ_i 's satisfy
 - $E(\epsilon_i) = 0$ for $i = 1, \dots, n$
 - $Var(\epsilon_i) = \sigma^2$ for $i = 1, \dots, n$
 - $Cov(\epsilon_i, \epsilon_j) = 0$ for all $i \neq j$
- Equivalently, the assumptions can be written as
 - $E(\epsilon) = \mathbf{0}$
 - $Var(\epsilon) = \sigma^2 I_n$

Estimation of Regression Coefficients:

- Consider the objective function

$$\begin{aligned} Q(\beta) &= Q(\beta_0, \beta_1, \dots, \beta_p) \\ &= \sum_{i=1}^n \left(y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip} \right)^2 \\ &= (y - X\beta)^T (y - X\beta) \\ &= (y^T y - 2\beta^T X^T y + \beta^T X^T X \beta) \end{aligned}$$

- The LSEs of $\beta_0, \beta_1, \dots, \beta_p$ are $\underset{\beta}{\operatorname{argmin}} Q(\beta)$.

Vector Derivative : Linear Function

- Let x be an $n \times 1$ column vector (the variable).
- Let a be a constant $n \times 1$ column vector.
- The derivative of a scalar-valued linear function with respect to the vector x is

$$\frac{\partial}{\partial x}(a^T x) = a.$$

Vector Derivative: Quadratic Function

- Let A be a constant $n \times n$ matrix.
- The derivative of a quadratic form with respect to the vector x is

$$\frac{\partial}{\partial x} (x^T A x) = (A + A^T) x.$$

- If A is a symmetric matrix, then the above formula simplifies to

$$\frac{\partial}{\partial x} (x^T A x) = 2Ax.$$

Estimation of Model Parameters:

- Differentiating $Q(\beta)$ with respect to β and setting the derivative to zero gives the following *normal equations*:

$$X^T X \beta = X^T y.$$

- The above system of linear equations in β is always consistent.
- If the matrix $X^T X$ is *invertible* (i.e. if X is of *full rank*), then the LSE is unique and given by

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

Fitted Value:

- The fitted value of response corresponding to regression $x = (1, x_1, \dots, x_p)$ is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p$$

- Then $\hat{\mathbf{y}} = (\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n)^T = X\hat{\beta} = X(X^T X)^{-1} X^T \mathbf{y} = H\mathbf{y}$, where, $H = X(X^T X)^{-1} X^T$ is called **hat-matrix**.
- The residuals:

$$\mathbf{e} = \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix} = \begin{bmatrix} y_1 - \hat{y}_1 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix} = \mathbf{y} - \hat{\mathbf{y}} = (I - H)\mathbf{y}$$

Properties of LSE of β

- $\hat{\beta}$ is a linear function of y
- $\hat{\beta}$ is unbiased estimator of β . That is, $E(\hat{\beta}) = \beta$
- $Var(\hat{\beta}) = \sigma^2(X^T X)^{-1}$
- $\hat{\beta}$ is the BLUE of β .

Estimation of Error Variance (σ^2)

- It can be shown that

$$E(SS_{Res}) = (n - p - 1)\sigma^2$$

- Hence, $\hat{\sigma}^2 = \frac{SS_{Res}}{n-p-1} = MS_{Res}$ is an unbiased estimator of σ^2 . Here MS_{Res} is residual mean square.
- Observed value of $\hat{\sigma}^2 = \frac{SS_{Res}}{n-p-1}$ is called **Residual variance**. It's **square root is called residual standard error**.
- $\frac{(n-p-1)MS_{Res}}{\sigma^2} \sim \chi^2_{n-p-1}$.
- Here,

$$SS_{Res} = \sum_{i=1}^n e_i^2 = y^T y - \hat{\beta}_1^T X^T y,$$

Hypothesis Testing: Test for Significance of Regression

- Assumption: $\epsilon \sim N_n(0, \sigma^2 I_n)$.
- Want to test the hypothesis if there is a linear relationship between the response y and any of the regressors $x_1, \dots, x_n$:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0 \text{ ag. } H_1 : \beta_j \neq 0 \text{ for atleast one } j = 1, \dots, p$$

- The test statistic is

$$F_0 = \frac{SS_{Reg}/p}{\hat{\sigma}^2} = \frac{SS_{Reg}/p}{SS_{Res}/(n-p-1)} \sim F_{p,n-p-1}, \text{ under } H_0.$$

- Reject H_0 iff $F_0 > F_{p,n-p-1;\alpha}$ (at level α)

Hypothesis Testing: Significance of regression coefficients

- Want to test

$$H_0 : \beta_j = 0 \text{ ag. } H_1 : \beta_j \neq 0$$

- Note that

$$\hat{\beta} \sim N_p \left(\beta, \sigma^2 (X^T X)^{-1} \right) \implies \hat{\beta}_j \sim N \left(\beta_j, \sigma^2 C_{jj} \right),$$

where C_{jj} is the j -th diagonal element of $(X^T X)^{-1}$

- The test statistic is

$$t_0 = \frac{\hat{\beta}_j}{\sqrt{MS_{Res} C_{jj}}} \sim t_{n-p-1} \text{ under } H_0$$

- Reject H_0 iff $|t_0| > t_{n-p-1, \alpha/2}$ (at level α)

Test of contribution of a subset of the regressors



$$y = X\beta + \epsilon \implies y = (X_1, X_2) \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + \epsilon$$

- Want to test:

$$H_0 : \beta_2 = 0 \text{ ag. } H_1 : \beta_2 \neq 0$$

- Based on the full model ($y = X\beta + \epsilon$), $\hat{\beta} = (X^T X)^{-1} X^T y$ and $SS_{Reg}(\beta)$ has p degrees of freedom.
- To find the contribution of β_2 , fit the model assuming $\beta_2 = 0$.
- The reduced model is $y = X_1\beta_1 + \epsilon$, $\hat{\beta}_1 = (X_1^T X_1)^{-1} X_1^T y$ and $SS_{Reg}(\beta_1)$ has $p - r$ degrees of freedom. Where r denotes the number of components in β_2

Test of contribution of a subset of the regressors

- $SS_{Reg}(\beta_2 \mid \beta_1) = SS_{Reg}(\beta) - SS_{Reg}(\beta_1)$ can be used as a measure of contribution of β_2
- Note that if β_2 has significant contribution then $SS_{Reg}(\beta_2 \mid \beta_1)$ is large
- Therefore, the test statistic is

$$F_0 = \frac{SS_{Reg}(\beta_2 \mid \beta_1)/r}{\hat{\sigma}^2} = \frac{SS_{Reg}(\beta_2 \mid \beta_1)/r}{SS_{Res}/(n-p-1)} \sim F_{r,n-p-1}, \text{ under } H_0$$

- Reject H_0 iff $F_0 > F_{r,n-p-1;\alpha}$ (at level α)

Confidence intervals for β_j :

- Confidence Interval of individual regression coefficient $\hat{\beta}_j$

- Pivot is

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{MS_{Res}C_{jj}}} \sim t_{n-p-1},$$

where C_{jj} is the diagonal element of $(X^T X)^{-1}$ matrix.

- A $100(1 - \alpha)\%$ CI for β_j is

$$\left[\hat{\beta}_j \pm t_{n-p-1, \alpha/2} \sqrt{MS_{Res}C_{jj}} \right].$$

Confidence interval for mean response:

- Let $x_0 = \begin{pmatrix} 1 \\ x_{01} \\ \vdots \\ x_{0p} \end{pmatrix}$ be a value of the regressor vector
- The mean response at x_0 is $x_0^T \beta$
- Pivot is $\frac{\hat{y}_0 - x_0^T \beta}{\sqrt{MS_{Res} x_0^T (X^T X)^{-1} x_0}} \sim t_{n-p-1}$
- A $100(1 - \alpha)\%$ CI for $x_0^T \beta$ is

$$\left[\hat{y}_0 \pm t_{n-p-1, \alpha/2} \sqrt{MS_{Res} x_0^T (X^T X)^{-1} x_0} \right]$$

Prediction Interval for New Observation

- Consider a level of regressor x_0 .
- Let y_0 be the corresponding value of the response.
- We want a prediction interval for y_0 .
- Let $\Psi = y_0 - \hat{y}_0 = y_0 - x_0^T \hat{\beta} \sim N(0, \sigma^2(1 + x_0^T (X'X)^{-1} x_0))$.
- A pivot is $\frac{\Psi - 0}{\sqrt{MS_{Res}(1 + x_0^T (X'X)^{-1} x_0)}} \sim t_{n-p-1}$.
- A $100(1 - \alpha)\%$ prediction interval of y_0 is

$$\left[\hat{y}_0 \mp t_{n-p-1; \frac{\alpha}{2}} \sqrt{MS_{Res}(1 + x_0^T (X'X)^{-1} x_0)} \right]$$

Standardized Regression Coefficients

- Difficult to compare regression coefficients. The magnitude of β_j reflects the unit of measurement of regressor x_j .
- For example, $y = 5 + x_1 + 1000x_2$, where y is measured in liters, x_1 in milliliters, and x_2 in liters. Here, $\beta_2 \gg \beta_1$. But the effects of both regressor on y are identical.
- Wayout is to standardized the regressors and response so that they become unit free.

Standardized Regression Coefficients

- A popular approach is as follows:

- Define $W_{ij} = \frac{x_{ij} - \bar{x}_j}{\sqrt{S_{jj}}}$, $i = 1, 2, \dots, n$; $j = 1, 2, \dots, p$.

$$y_i^* = \frac{y_i - \bar{y}}{\sqrt{SS_T}}, i = 1, 2, \dots, n.$$

$$\text{Here } S_{jj} = \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2, j = 1, 2, \dots, p.$$

- $\bar{W}_j = 0$ and $\sqrt{\sum_{i=1}^n (W_{ij} - \bar{W}_j)^2} = \sqrt{\sum_{i=1}^n W_{ij}^2} = 1$
 $\bar{y}^* = 0$ and $\sqrt{\sum_{i=1}^n (y_i^*)^2} = 1$

- In terms of y^* , $W_1, \dots, W_p$, the regression model becomes,

$$y^* = b_1 W_1 + b_2 W_2 + \dots + b_p W_p + \epsilon$$

- In matrix notation: $y^* = Wb + \epsilon$.
 - LSE: $\hat{b} = (W'W)^{-1}W'y^*$.

Standardized Regression Coefficients

- $W'W = \begin{pmatrix} 1 & r_{12} & \cdot & \cdot & \cdot & r_{1p} \\ r_{21} & 1 & \cdot & \cdot & \cdot & r_{2p} \\ \cdot & \cdot & & & & \cdot \\ \cdot & \cdot & & & & \cdot \\ \cdot & \cdot & & & & \cdot \\ r_{p1} & r_{p2} & \cdot & \cdot & \cdot & 1 \end{pmatrix}_{p \times p}$ $W'y^* = \begin{pmatrix} r_{1y} \\ r_{2y} \\ \cdot \\ \cdot \\ \cdot \\ r_{py} \end{pmatrix}_{p \times 1}$

- $r_{ij} = \frac{\sum_{u=1}^n (x_{ui} - \bar{x}_i)(x_{uj} - \bar{x}_j)}{\sqrt{S_{jj}S_{ii}}} = \frac{S_{ij}}{\sqrt{S_{jj}S_{ii}}}$

- $r_{iy} = \frac{\sum_{u=1}^n (x_{ui} - \bar{x}_i)(y_u - \bar{y})}{\sqrt{S_{jj}S_{ii}}} = \frac{S_{jy}}{\sqrt{S_{jj}SS_T}}$

Model Adequacy Checking (Regression Diagnostics)

- Major assumptions
 - linear relationship
 - Error mean zero
 - Error variance is constant
 - Error are uncorrelated
 - Error are normally distributed
- Gross violation of the assumptions may lead to a totally different model with opposite conclusions.
- Unusual observations: Sometimes just a few observations do not fit the model. These few observations might change the choice and fit of the model.
- We perform the checking using residuals.

Different Residuals

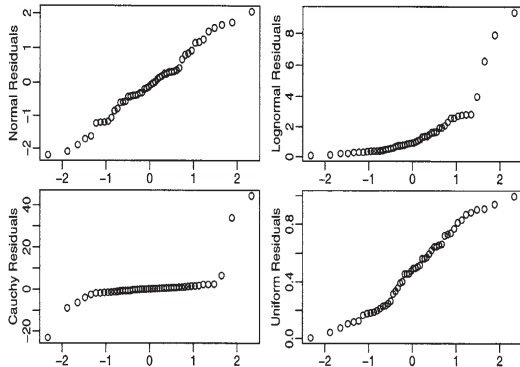
- We now define 3 types of residuals.
- Residual: $e_i = y_i - \hat{y}_i$.
- Standardized residual: $d_i = \frac{e_i}{\sqrt{MS_{Res}}}$
- Studentized residual:
 - $e = (I - H)\epsilon \Rightarrow Var(e) = \sigma^2(I - H)$.
 - $Var(e_i) = \sigma^2(1 - h_{ii})$ and $Cov(e_i, e_j) = -\sigma^2 h_{ij}$.
 - It can be shown that $0 \leq h_{ii} \leq 1$.
 - $r_i = \frac{e_i}{\sqrt{MS_{Res}(1 - h_{ii})}}$.

Residual Plots: Q-Q Plot (Test for normality)

- The residuals can be assessed for normality using a Q–Q plot. This compares the residuals to “ideal” normal observations.

- We plot the quantiles corresponding to sorted residuals (e_i) against $\Phi^{-1}(\frac{i}{n+1})$ for $i = 1, \dots, n$.

- Four Q-Q plots are:
Normal: ideal;
Lognormal: an example of a skewed distribution;
Cauchy: an example of a long-tailed (platykurtic) distribution;



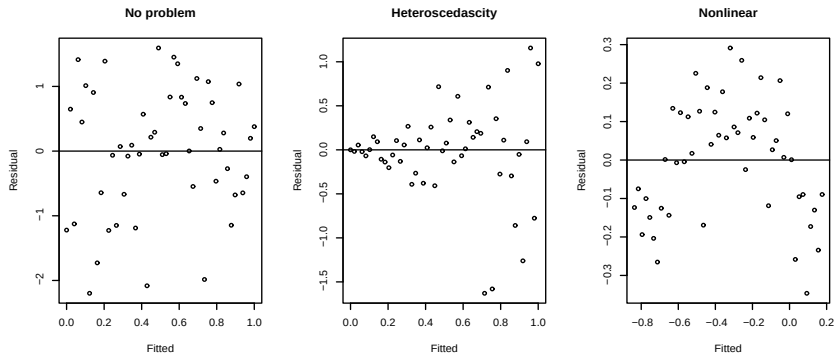
- **Uniform:** an example of a short-tailed (leptokurtic) distribution
- Source (book): Linear Models with R by Julian J. Faraway

If non-Normal?

- For short-tailed distributions, the consequences of nonnormality are not serious and can reasonably be ignored.
- For skewed errors, a transformation of the response may solve the problem.
- For heavy-tailed errors, we might just accept the nonnormality and base the inference on the assumption of another distribution or use resampling methods such as the bootstrap.

Residual Plots: Plot of residual against Fitted values

- Plot $\hat{y}_i$ vs e_i (or d_i or r_i): (Test of constant variance and non-linear relation)



Residuals vs Fitted plots - the **first** suggests **no change** to the current model while the **second** shows **non-constant variance** and the **third** indicates **some nonlinearity** which should prompt some change in the structural form of the model

Residual Plots

- Plot of residual against regressor
 - Plot x_{ij} vs $e_i \forall j$
 - In previous plot, replace 'fitted' by x_{ij} to find similar interpretation.
 - Repeat it for all the regressors.

If nonconstant variance?

- Two approaches
 - Weighted least squares
 - Transformation of variables
- Let h be a transformation that is applied on response

$$\begin{aligned}h(y) &= h(Ey) + (y - Ey)h'(Ey) + \dots \\ \text{var } h(y) &= h'(Ey)^2 \text{var } y + \dots\end{aligned}$$

We ignore the higher order terms. For $\text{var } h(y)$ to be constant we need

$$h'(Ey) \propto (\text{var } y)^{-1/2},$$

which suggests

$$h(y) = \int \frac{dy}{\sqrt{\text{var } y}} = \int \frac{dy}{\text{SD}y}.$$

- If $\text{var } y = \text{var } \varepsilon \propto (Ey)^2$, then $h(y) = \log y$ is suggested.
- If $\text{var } \varepsilon \propto (Ey)$, then $h(y) = \sqrt{y}$.

Checking Correlated Errors

- Index plot: Plot residual against index (time).
- Plot e_i against e_{i-1} .
- Remedy: Use generalized least squares method

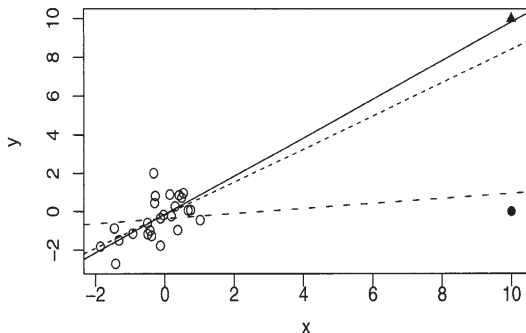
Checking the Structure of the Model

- How do we check whether the systematic part ($Ey = X\beta$) of the model is correct?
- Partial Regression Plot:
 - Complete/correct marginal effect
 - Let we want to study the marginal effect of x_j on y
 - y is regressed on $x_1, \dots, x_k$ except x_j .
 - x_j is regressed on $x_1, \dots, x_k$ except x_j .
 - Plot y residual, $e_i(y|x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_k)$ against x_j residual, $e_i(x_j|x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_k)$
 - For ideal scenario, the partial regression plot should show a linear relationship (straight line with non-zero slope).
 - Curvilinear band: indicates higher order terms in x_j or it's transformation.
 - Horizontal band: indicates no additional useful information in x_j
- Remedy: Change you model and/or add other regressors.

Finding Unusual Observations

- Outliers: Observations that do not fit the model well.
- Influential observations: Observations that change the fit of the model in a substantive manner.
- A point can be none, one or both of these.
- Leverage points: Observations that are unusual in the predictor space. It also has the potential to influence the fit.

Unusual Observations



- The solid line is the fit including the $\blacktriangle$ point but not the $\bullet$ point.
- The dotted line is the fit without both $\blacktriangle$ and $\bullet$ points.
- The dashed line is the fit with the $\bullet$ point but not the $\blacktriangle$ point.

Finding Outliers

- Order scaled residuals :

$$\frac{|e_i|}{\sqrt{\sum_{i=1}^n e_i^2}}$$

- The maximum one considerable larger (3 or 4 times) than others indicated potential outlier in y -space

Some points about outliers

- Two or more outliers next to each other can hide each other.
- An outlier in one model may not be an outlier in another when the variables have been changed or transformed.
- The error distribution may not be normal and so larger residuals may be expected.
- Individual outliers are usually much less of a problem in larger datasets.

Remedy

- Check for a data-entry error first. These are relatively common. Unfortunately, the original source of the data may have been lost.
- Examine the physical context—why did it happen?
- Exclude the point from the analysis but try reincluding it later if the model is changed.
- Suppose you find outliers that cannot reasonably be identified as mistakes or aberrations, but are viewed as naturally occurring – Use robust estimation
- It is dangerous to exclude outliers in an automatic manner.

Leverage

- h_{ii} : the i -th diagonal of hat matrix H
- h_{ii} 's are called leverages
- Note that

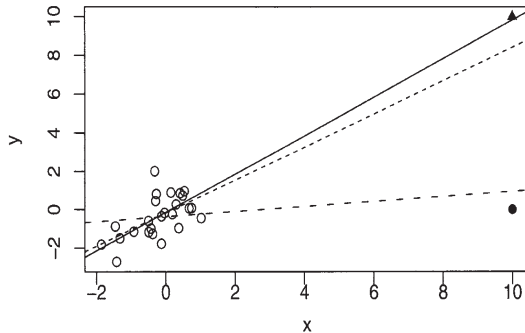
$$h_{ii} = x_i' (X'X)^{-1} x_i = \frac{1}{n} + \frac{MD_i^2}{n-1}$$

- $MD_i^2 = (x_i - \bar{x})^T S^{-1} (x_i - \bar{x})$: Mahalanobis distance
- Large values of h_{ii} indicates that the i -th observation is a leverage point

Leverage (continued)

- $\sum_{i=1}^n h_{ii} = p$ (as $\text{trace}(H) = \text{rank}(H) = \text{rank}(X) = p$)
- Average size of diagonals is $\frac{p}{n}$
- $h_{ii} > \frac{2p}{n}$: i -th observation is assumed to be leverage point
- Not all leverage points are going to be influential

Finding influential observations



Finding influential observations

- Cook's D :

$$D_i = \frac{(\hat{\beta} - \hat{\beta}_{(-i)})'(X'X)(\hat{\beta} - \hat{\beta}_{(-i)})}{p \cdot \text{MSE}} = \frac{\sum_{j=1}^n (\hat{y}_j - \hat{y}_{j(-i)})^2}{p \cdot \text{MSE}} = \frac{r_i^2}{p} \times \frac{h_{ii}}{1 - h_{ii}}$$

- $\hat{\beta}_{(-i)}$: LSE of β obtained by deleting the i -th observation
- $\hat{y}_{j(-i)}$: Fitted value of y_i obtained by deleting the i -th observation
- r_i : i -th Studentized residual
- Rule of thumb: $D_i > 1$ indicates i -th point is influential point

The PRESS Statistic

- How do you measure prediction capability of a fitted model?
- Delete i^{th} observation. Fit the model on remaining $(n - 1)$ observations. Now, predict y_i . Let the corresponding prediction error be $e_{(i)} = y - \hat{y}_{(i)}$ (PRESS Residual).
- $e_{(i)} = \frac{e_i}{1-h_{ii}}$. (Prove it by yourself)
- $Var(e_{(i)}) = \frac{\sigma^2}{1-h_{ii}}$.
- Standardized PRESS residual is $\frac{e_{(i)}}{\sqrt{Var(e_{(i)})}} = \frac{e_i}{\sqrt{\sigma^2(1-h_{ii})}}$, which is standardized residual if we estimate σ^2 by MS_{Res} .

The PRESS Statistic (continued)

- $PRESS = \sum_{i=1}^n e_{(i)}^2 = \sum_{i=1}^n \left(\frac{e_i}{1-h_{ii}} \right)^2$.
- PRESS is a measure of how well a regression model perform in predicting new observations.
- R^2 for prediction :
 $R_{prediction}^2 = 1 - \frac{PRESS}{SST}$: gives indication of the prediction capability of the regression model.
- Using PRESS, we may compare model.

Adjusted R^2

- In MLR, adding a variable to a model can increase the value of R^2 .
- To overcome this problem:

$$\text{Adjusted } R^2 : \quad R_{adj}^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2 / (n - p - 1)}{\sum_i (y_i - \bar{y})^2 / (n - 1)}.$$

Variable Selection

- Criteria for Evaluating Subset Regression Models:

- R^2
- R_{Adj}^2
- Residual Mean Square : $MS_{Res}(p) \equiv R_{Adj}^2$.
- Partial Correlation
- PRESS Statistic

- Techniques:

- All possible Regression.
- Step-wise Type Procedures :
 - Forward Selection : $F = \frac{SS_R(x_2|x_1)}{MS_{Res}(x_1, x_2)}$
 - Backward elimination
 - Step-wise Regression

Multicollinearity

- Near linear relationship among regressors ($\sum_{j=1}^p t_j \mathbf{x}_j \simeq 0$)
- Effect of Multicollinearity:
 - Consider scaled response and regressor (length unit).
 - Consider $y = \beta_1 x_1 + \beta_2 x_2 + \epsilon$.
 - $\hat{\beta}_1 = \frac{r_{1y} - r_{12}r_{2y}}{1 - r_{12}^2}$, and $\hat{\beta}_2 = \frac{r_{2y} - r_{12}r_{1y}}{1 - r_{12}^2}$.
 - $Var(\hat{\beta}_j) = \frac{\sigma^2}{1 - r_{12}^2}$, $Cov(\hat{\beta}_1, \hat{\beta}_2) = \frac{-r_{12}\sigma^2}{1 - r_{12}^2}$.
 - If $|r_{12}| \rightarrow 1$, $Var(\hat{\beta}_j) \rightarrow \infty$, and $|Cov(\hat{\beta}_1, \hat{\beta}_2)| \rightarrow \infty$.

- Effect of Multicollinearity (contd.):

- $L_1^2 = (\hat{\beta} - \beta)^T (\hat{\beta} - \beta).$
- $E(L_1^2) = \sum_{j=1}^p \text{Var}(\hat{\beta}_j) = \sigma^2 \text{Tr}(X'X)^{-1} = \sigma^2 \sum_{j=1}^p \frac{1}{\lambda_j},$ where λ_j 's are eigenvalues of $(X'X).$
- If $(X'X)$ is ill-conditioned then at least one λ_j will be small $\Rightarrow E(L_1^2)$ is big.
- Therefore, we have $E(\hat{\beta}^T \hat{\beta}) = \beta' \beta + \sigma^2 \text{Tr}(X'X)^{-1},$ implies magnitude of $\hat{\beta}$ are large.

Multicollinearity Diagnostics

- Examination of correlation matrix ($X'X$):
 - If x_i and x_j are nearly linearly dependent, then $|r_{ij}|$ should be close to 1.
 - However, this procedure is helpful to detect near linear dependence between a pair of regressors only.

Multicollinearity Diagnostics (Contd.)

- Variance Inflation Factors (VIFs):

- It can be shown that

$$Var(\hat{\beta}_j) = C_{jj} = (1 - R_j^2)^{-1} = \frac{1}{1 - R_j^2},$$

where R_j^2 is the coefficient of determination obtained when x_j is regressed on remaining $(k - 1)$ regressors.

- $VIF_j = C_{jj}$: This measures the factor by which the variance of $\hat{\beta}_j$ inflated due to the near linear dependence.
 - Rule of thumb : If any of $VIF > 5$, the associated coefficient is estimated poorly due to multicollinearity.

Multicollinearity Diagnostics (Contd.)

- Eigen System Analysis of $(X'X)$:
 - The eigen values, $\lambda_1, \lambda_2, \dots, \lambda_p$, can be used to see the extent of multicollinearity.
 - Small eigen values (one or more) $\Rightarrow$ multicollinearity.
 - Condition number $= \kappa = \frac{\lambda_{max}}{\lambda_{min}}$.
 - Rule of thumb :
 - $\kappa < 100 \rightarrow$ No serious problem with multicollinearity.
 - $100 \leq \kappa < 1000 \rightarrow$ moderate to strong multicollinearity.
 - $\kappa \geq 1000 \rightarrow$ severe multicollinearity.
 - Condition indices : $\kappa_j = \frac{\lambda_{max}}{\lambda_j}$, $j = 1, 2, \dots, p$
 - $\kappa_j \geq 1000 \rightarrow$ provide useful information on the number of near linear dependence.

Singular Value Decomposition (SVD)

Let $\mathbf{X}$ be the $n \times p$ matrix of regressors (usually centered and scaled). The SVD of $\mathbf{X}$ is:

$$\mathbf{X} = \mathbf{U}\mathbf{\Gamma}\mathbf{V}^T$$

where:

- $\mathbf{U}$ is an $n \times p$ matrix such that $\mathbf{U}^T\mathbf{U} = \mathbf{I}_p$.
- $\mathbf{V}$ is a $p \times p$ orthogonal matrix ($\mathbf{V}^T\mathbf{V} = \mathbf{I}_p$). The columns of $\mathbf{V}$ are the eigenvectors of $\mathbf{X}^T\mathbf{X}$.
- $\mathbf{\Gamma}$ is a $p \times p$ diagonal matrix of **singular values** $\lambda_1, \lambda_2, \dots, \lambda_p$.

Note: The eigenvalues of $\mathbf{X}^T\mathbf{X}$ are $\mu_j = \lambda_j^2$.

Condition Number and Condition Indices

The **Condition Number** of the matrix $\mathbf{X}$ is defined as:

$$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}}$$

The **Condition Indices** of the matrix $\mathbf{X}$ as:

$$\eta_j = \frac{\lambda_{\max}}{\lambda_j}, \quad j = 1, 2, \dots, p$$

Thresholds for η_j

- $\eta_j < 30$: Weak dependency.
- $30 < \eta_j < 100$: Moderate to strong dependency.
- $\eta_j > 100$: Severe multicollinearity.

Variance Decomposition Proportions

The covariance matrix of $\hat{\beta}$ is $\text{Var}(\hat{\beta}) = \sigma^2(\mathbf{X}^T \mathbf{X})^{-1}$. Using SVD:

$$\text{Var}(\hat{\beta}_j) = \sigma^2 \sum_{k=1}^p \frac{v_{jk}^2}{\lambda_k^2}$$

The **Variance Decomposition Proportion** (VDP) for the j -th regression coefficient associated with the k -th component is:

$$\pi_{kj} = \frac{v_{jk}^2 / \lambda_k^2}{\sum_{i=1}^p v_{ji}^2 / \lambda_i^2}$$

- π_{kj} represents the proportion of the variance of $\hat{\beta}_j$ explained by the k -th singular value.

Diagnostic Rule

A multicollinearity problem is diagnosed when:

- 1 A condition index η_k is large (typically > 30).
- 2 Two or more regressors have large variance decomposition proportions π_{kj} (typically > 0.5) associated with that large η_k .

Method for dealing with multicollinearity

- Source of multicollinearity:
 - Data collection method → collecting more data.
 - Constraints in model or population → Model respecification
 - Model specification → Model respecification
 - An overdefined model → Model respecification, and other method of estimate (like Ridge regression).

How to attack the data analysis/model fitting

- Understand the **research question(s)**. Understand the **data** you have.

- 1 How the **data was collected**?
- 2 What type of the **study design used**: *Randomized or Observational; Prospective or Retrospective etc.*
- 3 Can you make **connection with the primary research question and the data**? Is the **research question feasible** based on the data you have?
- 4 Are there **secondary research questions**?
- 5 What are the **potential source of bias**? Sample (data) **may not be** a representative of the target (source) population.
- 6 Are there any **confounders**?
Confounders: A confounder (also known as confounding variable, confounding factor) is **a variable that influences both the dependent variable and independent variable**, causing a spurious association.
Confounding is a causal concept, and as such, cannot be described in terms of correlations or associations.
- 7 Are the **number of observations/individuals** in the data **sufficient**? Is it a **small data**?

How to attack the data analysis/model fitting

- Do the **scatter plot(s)**: response vs. input variable(s).
- **Fit the regression** model(s) (or other type of model(s)).
- **Interpret** the output from the fitted models:
 - ① Are all the results expected? Whether the **results go well with existing domain** (basic science) knowledge?
 - ② If not, what are the **reasons behind the aberration** from the expected results.
- **Check the diagnostics** for model assumptions. If you find problem, go back and correct (if you can) the chosen model; or, take decision about the outliers/influential points.
- Always remember: *“Essentially, all models are wrong, but some are useful” – Box*